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Course name: Web application design	**Teacher's name**: Luis Fernando Buelna Ugalde
Module: 1	**Activity**: Evidence 1
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Learning Outcome of Module 1

“Halcon” is a construction material distributor that requires a web application to automate its internal processes. After an interview, the client's needs are as follows:

- A web application that allows their customers to see the status of their orders from a main screen where the customer enters a **customer number**, and an **invoice number**. The information to be displayed is the status and, in case of having a status of "Delivered", show the photo of the evidence that it was delivered. The **statuses** are as follows:
    - **Ordered**: when a material is ordered, and the sales executive enters it into the system.
    - **In process:** when the order is in stock, and is being prepared to go on route or, failing that, when it is not in stock and must be purchased from an external supplier.
    - **In route**: when the order has been routed for distribution.
    - **Delivered:** when the order has been delivered to the customer's premises.
- The personnel working in the company can access an administrative Dashboard to carry out their respective activities. This requires the following:
    - That the system brings by default an administrative user, which will be able to register new users and assign roles to the users.
    - The roles are the departments (important: customers will not be able to register):
        - **Sales:** those in charge of taking orders from customers.
        - **Purchasing:** in case of not having any material, these are the users who manage the purchase of materials.
        - **Warehouse:** who manage the warehouse and prepare the orders for routing, they also inform Purchasing about non-existent or low stock materials.
        - **Route:** who oversee distributing orders to customers.
    - The life cycle of an order is respected:

1. A customer calls the company to place an order.
2. The salesperson takes the order and assigns a new entry to the platform, which must contain the following:
    - Consecutive **invoice number** to which the order will correspond.
    - **Name or company name** of the customer placing the order.

- **A unique customer number** to be assigned arbitrarily.
- **Fiscal data of the client** for the filling of the physical invoice that later will be sent by e-mail (the application will not send invoices, each order is linked with an invoice number, but a person of administration oversees making the invoices separately).
- **Date and time** of the order.
- **Delivery address** of the order.
- A field to enter any notes or extra information.

3. The default status that the order acquires after it is raised is “**Ordered**” at that time the order should be visible to all employees of the company.
4. A warehouse person should take care of the order and change its status to "**In process".** Once he/she has finished gathering the materials (internal warehouse or by arranging with purchasing the acquisition of the missing product), he/she changes the status to "**In route**" and, in turn, physically loads a unit together with the carrier.
5. **The person on the route must take a photo of the loaded unit and upload it to the platform** (consider that the upload photo option should only be visible to people in the Route department).
6. Upon delivery, the operator **must take another photo of the unloaded material and upload it to the platform** as evidence of delivery.
7. Once the material is delivered, the status changes to "Delivered".
    - A screen where **all orders are listed**, with the possibility to search for orders by **Invoice Number, Customer Number, Date or Status.**
    - When an order is accessed, it can be modified or deleted logically, it is not deleted from the database, but a status is set so that it is not displayed together with the other orders.
    - A screen where the deleted orders are displayed, with the possibility to edit and restore them.

Link to GitHub: https://github.com/Jacobo-Rodriguez/Project-for-Halcon

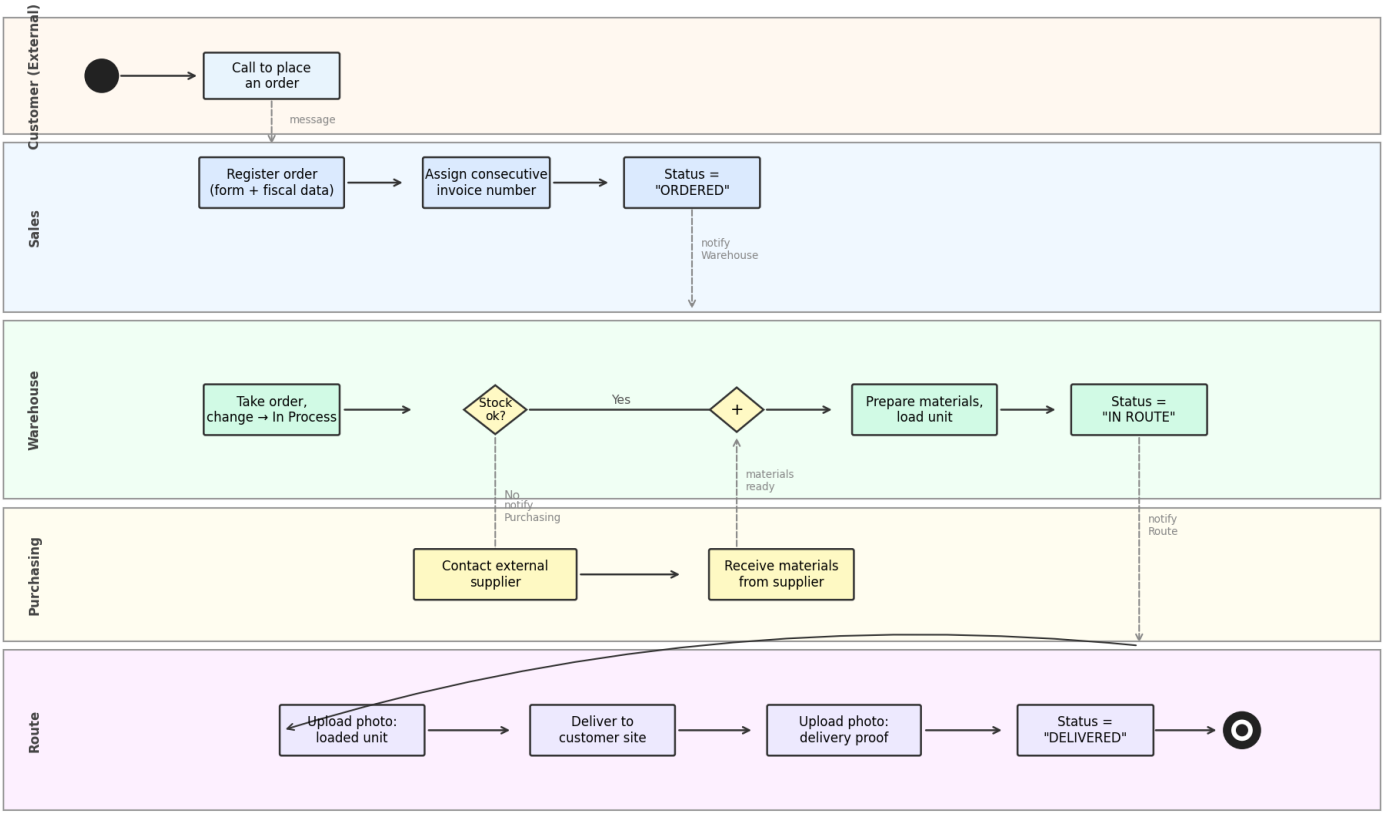
Choose and justify a work methodology that is ideal for the planning and execution of the project.

For the development of Halcon's web application, Scrum is proposed — the most widely used agile framework in the software industry. Scrum organizes work into short, iterative cycles called Sprints (typically two to four weeks), at the end of which a functional increment of the product is delivered.

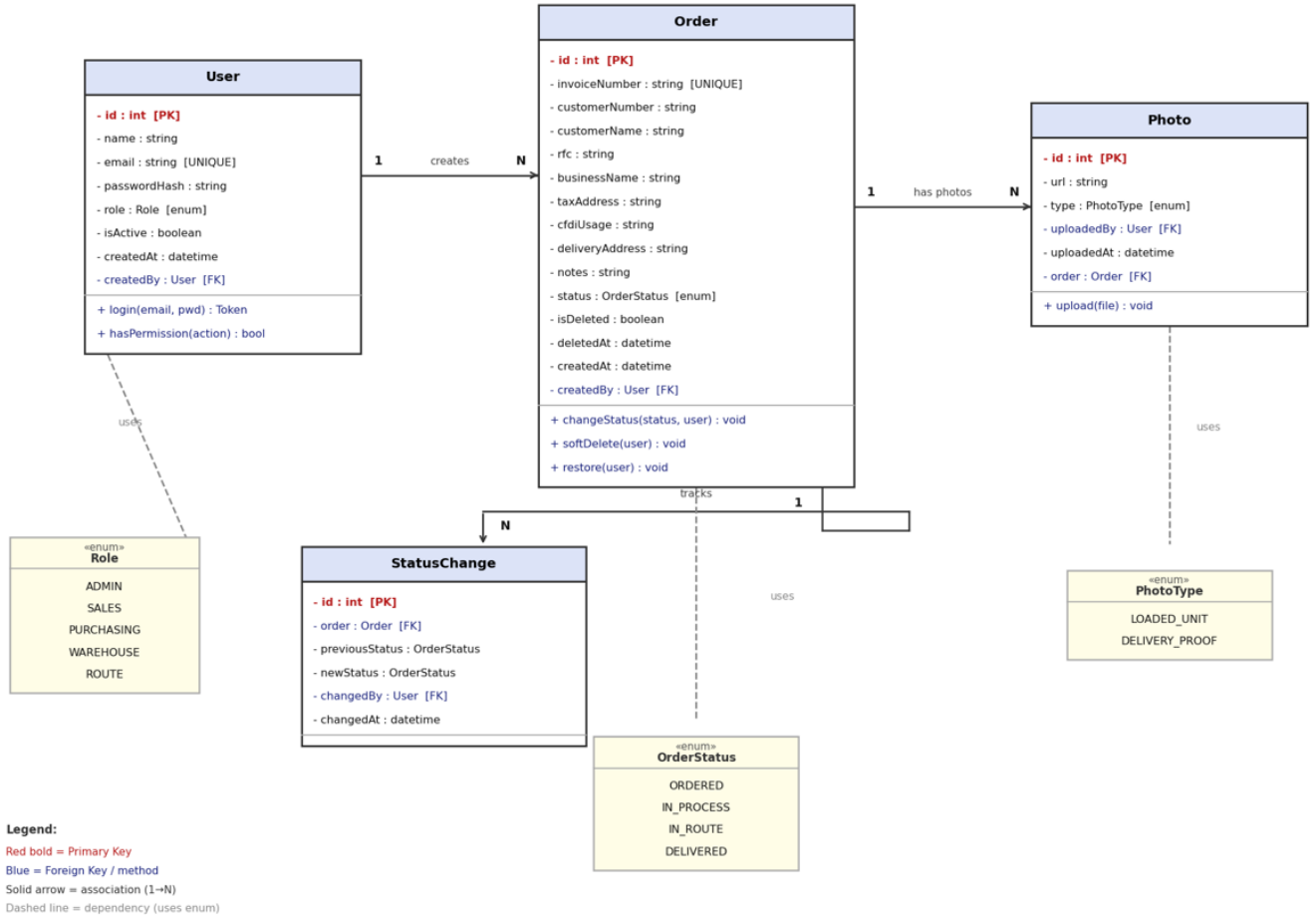
Scrum is a more fitting choice over alternatives like Waterfall, due to the evolving requirements of the client, especially on such an important and extensive piece of software, as well as prioritizing an early delivery, responsible risk management, saving up resources and continuous collaboration with the client.

Design the diagrams that will help you to describe graphically how the web application will behave. Among the diagrams that you should design, consider the following:

- **BPMN diagram**

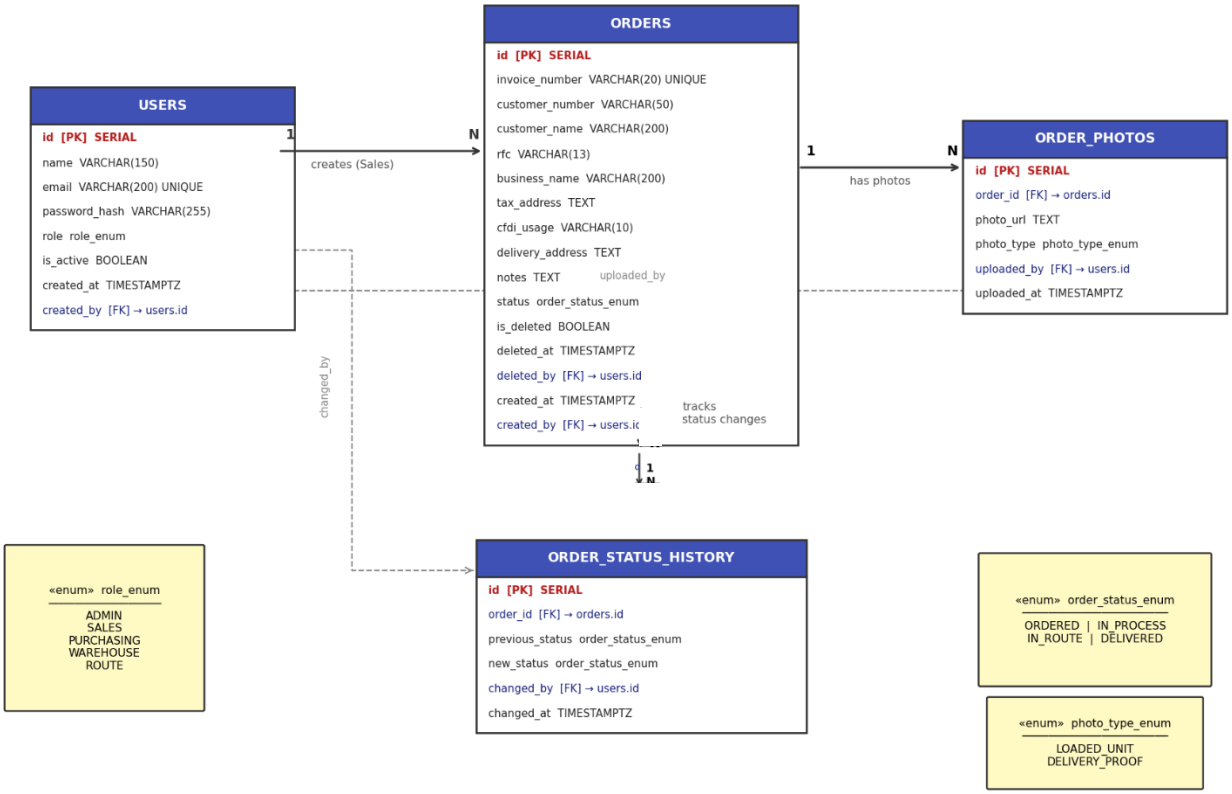


- **Class diagrams**



(Python suffered with this one)

Choose the database and its respective ER diagram containing all the entities with their attributes, and the way they are related.



Personal reflection.

Analyzing this system for Halcon turned out to be a particularly enriching exercise, since it demonstrated how business problems that appear simple on the surface reveal significant structural complexity when broken down with methodological approaches.

Some of the aspects I found interesting about the software planning were the separation of the customer portal from the administrative dashboard, based on their needs being fundamentally different.

The choice of Scrum as the methodology feels right due to the project's areas of uncertainty. For example, handling delivery photographs seems trivial during analysis, but in practice it involves important decisions about storage and related functionality.

This project demonstrates that good systems analysis is not purely technical, it is a careful translation between the natural practices and engineering. The more precise that translation, the more likely the final product will be genuinely useful to the people who rely on it.